

Muhammad Maaz Barki

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EDUCATION

Lahore University of Management Sciences Lahore, Punjab
BS Computer Science, Data Science 2022 – 2026
Relevant Coursework: Data Science, Intro to AI, Algorithms, Big Tech Foundations, Computer Vision, Causal Inference, Network Centric Computing, NLP, Azure Cloud Development.
Awards: Dean's Honor List 2023

EXPERIENCE

Lahore University of Management Sciences Lahore, Punjab
Head Teaching Assistant for Algorithms Fall 2025

- Facilitated a course of 300+ undergraduate and graduate students under Dr. Imdad Ullah Khan.
- Conducted weekly office hours to clarify complex algorithmic concepts (graph theory, dynamic programming, complexity theory) and graded assignments with constructive feedback.

Head Teaching Assistant for Business Data Management Spring 2026 – Present

- Leading a course of 60 students, instructing core data management including structured data, relational modeling, and SQL database design.
- Facilitate hands-on tutorials covering Python for data analysis (NumPy, Pandas, Matplotlib).
- Design technical assignments and E-R Modeling labs to enhance student mastery of data lifecycle concepts.

PROJECTS

Software Engineer - Lums Information Hub (LIH)  [Demo](#)

- Served as Technical Lead for full-stack requirements, architecting a robust MERN solution.
- Designed scalable backend MongoDB schemas and implemented RESTful API endpoints using Node.js/Express.
- Integrated Cloudinary for optimized media storage and utilized React for a responsive UI.
- Devised an automated testing strategy using Selenium for critical user flows (authentication, submission).

Technologies: MongoDB, Express.js, React.js, Node.js, Selenium, Git

Roman-Urdu Medical Chatbot  [Code](#)

- Developed a Speech-to-Text medical pipeline processing Urdu audio into Roman Urdu for health advisory.
- Built a RAG system using **LangChain** and **Sentence-Transformers** to mitigate LLM hallucinations.
- Optimized 1B/3B Llama and 7B Qwen models using **Unsloth** for memory-efficient training and faster inference.
- Evaluated performance using **BERTScore** to ensure diagnostic accuracy and high-fidelity text generation.

Technologies: LangChain, Unsloth, Transformers, Sentence-Transformers, BERTScore

3D Scene Reconstruction & Web Visualization Summer 2024  [Code](#)

- Developed an incremental Structure-from-Motion (SfM) pipeline to reconstruct detailed 3D environments from collections of 2D images using **OpenCV**.
- Implemented **SIFT** for robust feature detection and matching; utilized **EXIF** data and pairwise geometry to compute camera intrinsic and extrinsic parameter matrices.
- Executed incremental triangulation to build a 3D point cloud, applying **Global Bundle Adjustment** to minimize re-projection error and optimize spatial accuracy.
- Leveraged **Agisoft** for high-fidelity photogrammetry processing and integrated the final point cloud into an interactive web-based 3D viewer using **Three.js**.

Technologies: OpenCV, Agisoft, Three.js, Python, Photogrammetry, Computer Vision

RESEARCH EXPERIENCE

SDN-based UAV Networks: Congestion Prediction and Avoidance July 2025 – Present  [Reference](#)
Supervisor: Dr. Imdad Ullah Khan

- Investigating TCP/IP congestion control limitations in dynamic, high-latency SDN UAV environments.
- Developing simulation scenarios to evaluate protocol performance under varying UAV mobility patterns using RYU controllers.

SKILLS & INTERESTS

Languages: C/C++, Python (Expert), SQL, TypeScript, C#, Haskell, English (Fluent), Urdu (Native)
AI & Machine Learning: PyTorch, Scikit-learn, **LangChain**, **Unsloth**, Transformers, RAG, BERTScore, NLP
Data & Computer Vision: OpenCV, **Agisoft**, **Three.js**, Pandas, NumPy, Matplotlib, SIFT, SfM, Photogrammetry
Tools & Frameworks: MERN Stack (MongoDB, Express, React, Node), Git, **RYU (SDN Controller)**, Selenium, REST APIs, Cloudinary, Azure (In-Progress)
Interests: Fitness/Athleticism, Competitive Gaming, 3D Printing Technology, High-performance Computing